CANDIDATE
NAME

CLASS

ADMISSION
NUMBER

2020 Preliminary Examination Pre-University 3

MATHEMATICS

9758/02

Paper 2

17 Sep 2020

3 hours

Candidates answer on the Question Paper.

Additional Materials: List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your admission number, name and class on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give your answers in the spaces provided in the Question Paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

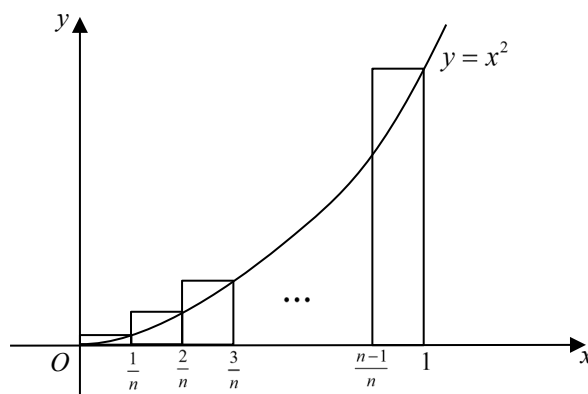
The total number of marks for this paper is 100.

Qn No.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	*	Total
Score													
Max Score	4	7	8	11	10	5	7	10	12	12	14		100

This document consists of **27** printed pages and **1** blank pages.

Section A: Pure Mathematics [40 marks]

1 (i)



The diagram shows the graph of $y = x^2$ and n rectangles, each with equal width, from $x = 0$ to $x = 1$. Show that the total area of the n rectangles can be written as

$$\frac{1}{n^3} \sum_{r=1}^n r^2. \quad [2]$$

(ii) Hence find $\lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{r=1}^n r^2$. [2]

2 It is given that

$$f(x) = \begin{cases} x^2 - 1 & \text{for } -2 \leq x \leq 2, \\ 3 & \text{for } 2 < x < 4, \end{cases}$$

and that $f(x+6) = f(x)$ for all real values of x .

(i) Sketch the graph of $y = f(x)$ for $-6 \leq x \leq 10$. [3]

(ii) On the same diagram in part **(i)**, sketch the graph of $y = 3 + \ln(x+2)$. [2]

(iii) Hence solve the inequality $f(x) \geq 3 + \ln(x+2)$. [2]

3 Do not use a calculator in answering this question.

- (a) (i) The roots of the equation $z^2 + 8z + 25 = 0$ are denoted by z_1 and z_2 , where $\arg(z_1) > 0$. Find z_1 and z_2 and show these roots on an Argand diagram.

[3]

- (ii) Find the exact modulus and argument of $z_1 + 1$.

[2]

- (b) For any two complex numbers w_1 and w_2 , the triangle inequality states that

$$|w_1| + |w_2| \geq |w_1 + w_2|.$$

By considering the complex numbers $1 + ic$ and $c + i$, where c is a real and positive constant, use the triangle inequality to prove that $c + 1 \leq \sqrt{2(1 + c^2)}$.

[3]

4 A curve C , known as a Lissajous curve, has parametric equations

$$x = a \sin(pt + q), \quad y = b \sin t,$$

where a, b, p and q are constants and $-\pi \leq t \leq \pi$.

- (a) Given that $p = 1$ and $q = \frac{\pi}{2}$,

- (i) show that $\frac{dy}{dx} = -\frac{b}{a} \cot t$.

[3]

- (ii) Determine the value of the parameter t at the point $\left(\frac{a\sqrt{3}}{2}, \frac{b}{2}\right)$.

[3]

- (iii) The normal to C at the point $\left(\frac{a\sqrt{3}}{2}, \frac{b}{2}\right)$ has equation $y = \frac{2}{\sqrt{3}}x - \frac{3}{2}$. Show that $a = 2$ and find the value of b .

[4]

- (b) When $p = 1$ and $q = 0$, show that C is a straight line.

[1]

5 Find

(i) $\int \frac{1}{1-4x^2} dx$, [2]

(ii) $\int \frac{x}{\sqrt{1-4x^2}} dx$. [2]

(iii) Using your answers to parts (i) and (ii), find $\int \left(2 - \frac{4x}{\sqrt{1-4x^2}}\right)^2 dx$. Hence find the exact area of the region bounded by the curve $y = \left(2 - \frac{4x}{\sqrt{1-4x^2}}\right)^2$, the x - and y -axes, and the line $x = \frac{1}{4}$. [6]

Section B: Probability and Statistics [60 marks]

6 Find the number of ways in which all the letters of the word FANTASTIC can be arranged if

(i) there are no restrictions, [2]

(ii) the first and last letters are consonants and the vowels are separated. [3]

7 Brandon and Kamala play a game, where Brandon throws a fair six-sided die and a biased four-sided die once. The fair six-sided die has the numbers '1' to '6' labelled on its faces and the biased four-sided die has the numbers '1' to '4' labelled on its faces. The score on each of the die is the number that is labelled on the face which lands on the table. The score on the biased four-sided die follows the following probability distribution:

$$P(X=r) = \begin{cases} kr^2 + \frac{1}{3} & , r=1, 2, \\ kr & , r=3, 4, \end{cases}$$

where k is a non-zero positive constant.

(i) Find the value of k . [2]

(ii) Brandon loses \$6 to Kamala if the score on the biased four-sided die is at least twice the score on the fair six-sided die. Both players receive nothing when the scores on both the dice are the same. Kamala loses \$4 to Brandon for all other outcomes. Let X be Brandon's winnings after one game. Determine the probability distribution of X . [3]

(iii) Find $E(X)$. Hence determine if the game is in favour of Brandon or Kamala in the long run. [2]

8 A group of n people each takes turn to pick one card, with replacement, from a pack of cards.

(a) Assume that there are 12 unique cards in a pack. The probability that all the people in the group picked different cards is denoted by P .

(i) For the case where $n = 3$, show that $P = \frac{55}{72}$. [2]

(ii) Find P for the case where $n = 4$ and deduce the smallest value of n such that $P < \frac{1}{2}$. [2]

(iii) State, with a reason, the least possible value of n such that $P = 0$. [2]

(b) Assume now that there are 120 unique cards in the pack. It is given that, for the case where $n = 12$, the probability that all the people in the group picked different cards is 0.56640, correct to 5 decimal places. Find the smallest value of n such that the probability that at least two people picked the same card exceeds $\frac{1}{2}$. [4]

9 Always Cool Company produces reusable ice pack gels. It is known that, on average, $p\%$ of the reusable ice pack gels leak. Each day the quality control department takes a random sample of 120 reusable ice pack gels and checks them for leakage. The number of reusable ice pack gels that leak in a sample is denoted by the random variable X .

(i) State, in the context of the question, two assumptions needed for X to be well-modelled by a binomial distribution. [2]

Assume now that X can be modelled by a binomial distribution.

- (ii) It is given that the most likely number of reusable ice pack gels that leak in a sample is 3. Find the range of values of p . [3]

For the rest of the question, the value of p is taken to be 3.

- (iii) Find the probability that the quality control department finds at most 4 reusable ice pack gels that leak in a sample. [1]
- (iv) If the quality control department found at least 5 reusable ice pack gels that leak in a sample, find the probability that there are fewer than 9 ice pack gels that leak in a sample. [3]
- (v) The number of reusable ice pack gels that leak in a sample is independent of that of other samples. Find the probability that, in 10 randomly chosen samples, the eighth sample is the third sample with at most 4 reusable ice pack gels that leak. [3]

10 In this question you should, where applicable, state clearly all the distributions that you use, together with the values of the appropriate parameter.

The mass, in grams, of a randomly chosen orange sold at Fantastic Supermarket follows the normal distribution $N(120, 16^2)$ and the mass, in grams, of a randomly chosen carrot sold at the same supermarket follows the normal distribution $N(73, 6^2)$.

- (i) Find the probability that the mass of a randomly chosen orange sold at Fantastic Supermarket lies within 5 grams of its mean mass. [2]
- (ii) Find the set of values of k such that less than 5% of the carrots sold at Fantastic Supermarket have a mass greater than k grams. [2]

A recipe for making carrot and orange juice requires 4 oranges and 7 carrots.

- (iii) Find the probability that the total mass of 4 randomly chosen oranges sold at Fantastic Supermarket exceeds the total mass of 7 randomly chosen carrots sold at Fantastic Supermarket by at least 5 grams. [3]

To make a cup of orange and carrot juice, the oranges and carrots must first be peeled and the seeds of the oranges removed. This preparation process reduces the mass of each orange by 30% and reduces the mass of each carrot by 12%.

- (iv) Find the probability that the total mass, after the preparation process, of 8 randomly chosen oranges and 14 randomly chosen carrots sold at Fantastic Supermarket is less than 1600 grams. [4]
- (v) State, in the context of the question, an assumption required in your calculations in parts (iii) and (iv). [1]
- 11 With the rising number of students owning a mobile device, Nurturing Primary School decides to conduct a survey on the amount of screen time the students spend on their mobile devices each day. There are 8 classes for each level from Primary One to Primary Six.
- (i) As Mrs Neha only teaches the Primary Six cohort, she decides to conduct the survey for all the 8 classes of Primary Six students. Explain whether these 8 classes of Primary Six students form a sample or a population. [1]
- (ii) For the data collected to be more representative of the entire student population, Mrs Neha decides to conduct the survey for 100 students from Primary One to Primary Six. Explain how she should select the 100 students and why she should do it in this way. [2]
- (iii) Based on national data, the mean screen time per day of students is found to be 30 minutes. Through her interactions with students in the school, Mrs Neha claims that the mean screen time per day should be at least 30 minutes. A random sample of 100 students from Nurturing Primary School is taken and their screen time per day is noted.

The screen time per day, x minutes, of the 100 students are summarised as follows.

$$\sum (x - 30) = -43 \quad \sum (x - 30)^2 = 911$$

- (a) Calculate unbiased estimates of the population mean and variance of the screen time per day of students at Nurturing Primary School. [3]
- (b) Test, at the 10% level of significance, if Mrs Neha's claim is justified. You should state your hypotheses and define any symbols you use. [5]
- (c) Another teacher, Ms Wu, takes another random sample of 100 students from Nurturing Primary School, and finds that they have a mean screen time per day of 30.3 minutes. In a test at the 5% level of significance, Ms Wu concludes that the mean screen time per day of students differs from 30 minutes. Find the range of possible values of the population variance used in calculating the test statistic. [3]

End of Paper